Topics in Error-Correcting Codes

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Lecture 13: Potentials, LCL Profiles, and Subspace Designs

Profiles and Their Potential

From now on, we use r for the locality and say *r-LCL profile*, replacing the L notation from Lecture 12. All dimensions, ranks, and linear maps are over $\mathbb{F}_q$ unless stated otherwise. We define containment directly for additive codes over vector symbols.

Definition 1 (r-LCL profile and containment)**.** An *r-local coordinate-wise linear profile* is a tuple of linear maps

$$(V_1, \dots, V_n), \qquad V_i : \mathbb{F}_q^r \longrightarrow \mathbb{F}_q^r.$$

An additive code $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ *contains* this profile if there are linearly independent codewords $c_1, \dots, c_r \in \mathcal{C}$ such that

$$V_i(c_{1,i}, \dots, c_{r,i}) = 0 \qquad (i \in [n]),$$

where V_i is applied separately to each of the s coordinates of these vector symbols.

Only the kernels of the maps matter for containment.

Definition 2 (Potential)**.** For a subspace $W \leq \mathbb{F}_q^r$ and a real parameter α, define

$$\Phi_W\big((V_1, \dots, V_n), \alpha\big) = \alpha \dim W - \frac{1}{n} \sum_{i=1}^{n} \big(\dim W - \dim(W \cap \ker V_i)\big).$$

The term $\dim W - \dim(W \cap \ker V_i)$ is the number of independent constraints imposed by V_i when its input is restricted to W. Negative potential means there are too many constraints: on average, more than $\alpha \dim W$ independent constraints on W. In particular, $\Phi_{\{0\}} = 0$.

Random Linear Codes and Quotients

For scalar symbols ($s = 1$), choose $G \in \mathbb{F}_q^{k \times n}$ uniformly at random and let

$$\mathcal{C} = \{uG : u \in \mathbb{F}_q^k\}, \qquad k = Rn.$$

For random-generator codes, R denotes the design rate; we do not condition on G having full row rank. Recall the first-moment estimate from Lecture 12.

Theorem 3 (Containment bound from Lecture 12)**.** *For every fixed r-LCL profile* $(V_1, \dots, V_n)$*,*

$$\Pr[\mathcal{C} \textit{ contains } (V_1, \dots, V_n)] \leq q^{n\Phi_{\mathbb{F}_q^r}((V_1,\dots,V_n),R)}.$$

Thus a whole-space potential at most $-\varepsilon$ gives containment probability at most $q^{-\varepsilon n}$. For a random additive code with a uniform generator $G \in \mathbb{F}_q^{k\times sn}$ and rate $R = k/(sn)$, the same bound has exponent $sn\Phi_{\mathbb{F}_q^r}((V_1,\dots,V_n),R)$: every local equation is imposed on all s scalar entries.

The whole-space potential need not be negative even when a profile is unlikely to be contained. To get a stronger test, we pass to linear images of its allowed spaces.

Surjective maps preserve containment

Let $\pi : \mathbb{F}_q^r \to \mathbb{F}_q^d$ be a surjective linear map, with $d \geq 1$. Choose linear maps $\widehat{V}_i : \mathbb{F}_q^d \to \mathbb{F}_q^d$ such that

$$\ker \widehat{V}_i = \pi(\ker V_i) \qquad (i \in [n]).$$

These maps specify the *image profile* by projecting the allowed spaces. The original V_i need not induce maps on the quotient.

Lemma 4 (Containment under a surjection)**.** *If an additive code $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ contains $(V_1,\dots,V_n)$, then it contains $(\widehat{V}_1,\dots,\widehat{V}_n)$.*

Proof. Let $c_1,\dots,c_r$ be independent codewords realizing the original profile. Write $\pi = (\ell_1,\dots,\ell_d)$, where $\ell_h(x_1,\dots,x_r) = \sum_{j=1}^r a_{hj}x_j$, and define

$$c'_h = \sum_{j=1}^r a_{hj}c_j \in \mathcal{C} \qquad (h \in [d]).$$

Surjectivity of π means that its component linear forms $\ell_1,\dots,\ell_d$ are independent. A relation among the c'_h would, by independence of the c_j, give the same relation among these forms. Thus $c'_1,\dots,c'_d$ are independent.

At each coordinate, $(c'_{1,i},\dots,c'_{d,i})$ is obtained by applying π coordinatewise to $(c_{1,i},\dots,c_{r,i})$. Since $\pi(\ker V_i) = \ker \widehat{V}_i$, the original containment equations give $\widehat{V}_i(c'_{1,i},\dots,c'_{d,i}) = 0$ coordinatewise. Thus the new codewords realize the image profile. □

The map must be the same at every coordinate: this is what makes the new words fixed linear combinations of codewords.

The potential of a quotient

Fix a proper subspace $W < \mathbb{F}_q^r$, put $d = r - \dim W$, and choose a surjection $\pi : \mathbb{F}_q^r \to \mathbb{F}_q^d$ with kernel W. This identifies $\mathbb{F}_q^d$ with $\mathbb{F}_q^r/W$. Rank-nullity on $\ker V_i$ gives

$$\dim \pi(\ker V_i) = \dim \ker V_i - \dim(W \cap \ker V_i).$$

Substituting this identity into the definition of the potential shows that, for every α,

$$\Phi_{\mathbb{F}_q^d}\big((\widehat{V}_1,\dots,\widehat{V}_n),\alpha\big) = \Phi_{\mathbb{F}_q^r}\big((V_1,\dots,V_n),\alpha\big) - \Phi_W\big((V_1,\dots,V_n),\alpha\big).$$

Corollary 5 (Containment bound for every subspace)**.** *Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a random linear code of rate R. For every $W \leq \mathbb{F}_q^r$,*

$$\Pr[\mathcal{C} \text{ contains } (V_1, \ldots, V_n)] \leq q^{n\left[\Phi_{\mathbb{F}_q^r}((V_1,\ldots,V_n),R) - \Phi_W((V_1,\ldots,V_n),R)\right]}.$$

For a random additive code in $(\mathbb{F}_q^s)^n$ of rate $R = k/(sn)$, replace n in the exponent by sn.

Proof. If $W = \mathbb{F}_q^r$, the right-hand side is 1. Otherwise use the quotient map above. Containment of the original profile implies containment of the image profile by Lemma 4. Apply Theorem 3 to the image profile and use the quotient-potential identity above. □

Taking $W = \{0\}$ recovers the bound from Lecture 12. A different W may have larger potential than the whole space, even when the whole-space potential is nonnegative. Then the quotient has negative potential, and the stronger bound rules out containment with high probability.

Take $r = 2$, $R = 1/2$, and $4 \mid n$. Require c_1 to vanish on $3n/4$ fixed coordinates and leave c_2 unrestricted. The whole-space potential is $1/4$. The subspace $W = \operatorname{span}\{(0,1)\}$ has potential $1/2$, so the quotient retaining only c_1 has potential $-1/4$. For scalar symbols, this gives containment probability at most $q^{-n/4}$.

The Containment Threshold

The reverse direction says that these quotient obstructions are complete for random linear codes, with quantitative slack.

Theorem 6 (Containment above threshold; Problem Set 3)**.** *Let $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$ be generated by a uniformly random $G \in \mathbb{F}_q^{k \times sn}$, and put $R = k/(sn)$. Suppose $\varepsilon > 0$ and, for every proper subspace $W < \mathbb{F}_q^r$,*

$$\Phi_{\mathbb{F}_q^r}\big((V_1, \ldots, V_n), R\big) \geq \Phi_W\big((V_1, \ldots, V_n), R\big) + \varepsilon.$$

Then

$$\Pr[\mathcal{C} \text{ contains } (V_1, \ldots, V_n)] \geq 1 - q^{r^2 - sn\varepsilon}.$$

We will not prove this direction in lecture; it is the content of Problem Set 3, Problem 3. For fixed r and positive ε, the failure bound tends to zero as n grows. The scalar statement is the case $s = 1$.

Definition 7 (Threshold of a profile)**.** The *containment threshold* $R_*(V_1, \ldots, V_n)$ is the smallest $\alpha \in [0, 1]$ such that

$$\Phi_{\mathbb{F}_q^r}\big((V_1, \ldots, V_n), \alpha\big) \geq \Phi_W\big((V_1, \ldots, V_n), \alpha\big) \qquad \text{for every } W < \mathbb{F}_q^r.$$

For each proper W, the difference between the whole-space and W potentials is affine in α, with positive slope $r - \dim W$. These differences are all nonnegative at $\alpha = 1$, so the defining set is nonempty and the minimum exists. At the threshold, at least one proper subspace ties the whole space. Above it, the whole space has strictly larger potential than every proper subspace.

Consequently, for a random additive code of rate R and any $\varepsilon > 0$,

$$R \leq R_*(V_1, \ldots, V_n) - \varepsilon \implies \Pr[\mathcal{C} \text{ contains } (V_1, \ldots, V_n)] \leq q^{-sn\varepsilon},$$
$$R \geq R_*(V_1, \ldots, V_n) + \varepsilon \implies \Pr[\mathcal{C} \text{ contains } (V_1, \ldots, V_n)] \geq 1 - q^{r^2 - sn\varepsilon}.$$

The first implication follows from the quotient bound; the second is the claim assigned in Problem Set 3. The positive slopes ensure that a rate gap of ε gives a potential gap of at least ε. We make no assertion about the probability exactly at the threshold.

Subspace Designs Give Universal Noncontainment

We now return to arbitrary additive codes $\mathcal{C} \subseteq (\mathbb{F}_q^s)^n$. Recall the definition from Lecture 12. For a linear subspace $\mathcal{A} \subseteq \mathcal{C}$, write

$$\mathcal{A}_i = \{a \in \mathcal{A} : a_i = 0\}.$$

Definition 8 (τ-subspace-design code)**.** Let $\tau : \mathbb{N} \to [0,1]$ be nondecreasing. The code $\mathcal{C}$ is a *τ-subspace-design code* if, for every $r \geq 1$ and every subspace $\mathcal{A} \subseteq \mathcal{C}$ of dimension at most r,

$$\frac{1}{n}\sum_{i=1}^{n} \dim \mathcal{A}_i \leq \tau(r) \dim \mathcal{A}.$$

For one-dimensional $\mathcal{A}$, this is a bound on the fraction of zero coordinates of a nonzero codeword. For larger $\mathcal{A}$, it bounds the average dimension lost under coordinate evaluation. In potential notation, it gives the following universal statement.

Proposition 9 (Universal noncontainment)**.** *If $\mathcal{C}$ is a τ-subspace-design code, it contains no r-LCL profile $(V_1, \ldots, V_n)$ for which*

$$\boxed{\Phi_{\mathbb{F}_q^r}\big((V_1, \ldots, V_n), \tau(r)\big) < 0.}$$

Proof. Suppose independent codewords $c_1, \ldots, c_r$ realize the profile, and let $\mathcal{A} = \operatorname{span}\{c_1, \ldots, c_r\}$. The coefficient map $a \mapsto \sum_{j=1}^{r} a_j c_j$ identifies $\mathbb{F}_q^r$ with $\mathcal{A}$, and identifies

$$H_i = \left\{ a \in \mathbb{F}_q^r : \sum_{j=1}^{r} a_j c_{j,i} = 0 \right\}$$

with $\mathcal{A}_i$. A component equation $\sum_j a_j x_j = 0$ of V_i gives $\sum_j a_j c_{j,i} = 0$ by containment. Thus $(a_1, \ldots, a_r) \in H_i$. These coefficient vectors span a space of dimension $r - \dim(\ker V_i)$. Hence $r - \dim(\ker V_i) \leq \dim H_i = \dim \mathcal{A}_i$, and

$$\frac{1}{n}\sum_i \big(r - \dim(\ker V_i)\big) \leq \frac{1}{n}\sum_i \dim \mathcal{A}_i \leq \tau(r) r.$$

By the definition of the potential, this says that $\Phi_{\mathbb{F}_q^r}((V_1, \ldots, V_n), \tau(r)) \geq 0$. □

Conversely, universal avoidance of these profiles at every locality implies the subspace-design property. Given a basis $c_1, \ldots, c_r$ of $\mathcal{A}$, choose the component linear forms of V_i so that their coefficient vectors span H_i above. Then these codewords realize the profile and $r - \dim(\ker V_i) = \dim \mathcal{A}_i$, so a violation of the design bound would give a contained profile of negative potential. Monotonicity of τ passes from dimension r to dimension at most r.

Unlike the random-code estimate for a profile fixed in advance, this exclusion holds *simultaneously for every profile*. There is no union-bound cost for the number of profiles.

Applying the universal bound to quotients

Suppose $W < \mathbb{F}_q^r$ and put $d = r - \dim W$. By preservation of containment and the quotient-potential identity, a τ-subspace-design code cannot contain $(V_1, \dots, V_n)$ if

$$\Phi_{\mathbb{F}_q^r}\big((V_1, \dots, V_n), \tau(d)\big) < \Phi_W\big((V_1, \dots, V_n), \tau(d)\big).$$

In particular,

$$\tau(r) < R_*(V_1, \dots, V_n) \quad \Longrightarrow \quad \mathcal{C} \text{ does not contain } (V_1, \dots, V_n).$$

Folded Reed–Solomon Codes Are Subspace Designs

Let $1 \le k \le sn$, let $\gamma \in \mathbb{F}_q^*$ have multiplicative order at least k, and suppose the sn evaluation points

$$\{\alpha_i \gamma^a : i \in [n],\ 0 \le a < s\}$$

are distinct and nonzero. Recall the s-folded Reed–Solomon code

$$f \longmapsto \big((f(\alpha_i), f(\alpha_i\gamma), \dots, f(\alpha_i\gamma^{s-1}))\big)_{i=1}^{n}, \qquad \deg f < k.$$

Evaluation is injective, so its dimension is k and its rate is $R = k/(sn)$.

For a subspace $\mathcal{A} \subseteq \mathcal{C}$, identify its codewords with their message polynomials. Then

$$\mathcal{A}_i = \{f \in \mathcal{A} : f(\alpha_i\gamma^a) = 0 \text{ for every } 0 \le a < s\}, \qquad r_i = \dim \mathcal{A}_i.$$

Thus $\mathcal{A}_i$ consists of the codewords in $\mathcal{A}$ whose entire ith folded symbol is zero.

Theorem 10 (FRS subspace-design bound)**.** *For every $1 \le r \le s$ and every subspace $\mathcal{A} \subseteq \mathcal{C}$ of dimension at most r,*

$$\frac{1}{n}\sum_{i=1}^{n} r_i \le \frac{sR}{s - r + 1}\dim \mathcal{A}.$$

Thus FRS is a τ-subspace-design code with

$$\tau(r) = \min\left\{1, \frac{sR}{s-r+1}\right\} \quad (1 \le r \le s), \qquad \tau(r) = 1 \quad (r > s).$$

The algebraic input is the full-rank criterion for the folded Wronskian. Its rank is taken over the rational-function field $\mathbb{F}_q(X)$, not at each individual evaluation point.

Lemma 11 (Folded Wronskian; Problem Set 3)**.** *Let $f_1, \dots, f_t \in \mathbb{F}_q[X]$ have degree less than k, where $1 \le t \le k$. If γ has multiplicative order at least k, the matrix*

$$\mathcal{W}_f(X) = \begin{pmatrix} f_1(X) & \cdots & f_t(X) \\ f_1(\gamma X) & \cdots & f_t(\gamma X) \\ \vdots & & \vdots \\ f_1(\gamma^{t-1}X) & \cdots & f_t(\gamma^{t-1}X) \end{pmatrix}$$

has rank t over $\mathbb{F}_q(X)$ if and only if $f_1, \dots, f_t$ are linearly independent over $\mathbb{F}_q$. Equivalently, $\det \mathcal{W}_f(X)$ is nonzero exactly when the polynomials are linearly independent.

The proof of this criterion is assigned in Problem Set 3, Problem 1(a). We use it without proof here.

Proof of Theorem 10. Let $t = \dim \mathcal{A} \le r$; the case $t = 0$ is immediate. Choose a basis $f_1, \dots, f_t$ of $\mathcal{A}$ and put $D(X) = \det \mathcal{W}_f(X)$. By the lemma, $D \neq 0$, and $\deg D \le t(k-1)$.

Fix i. Choose a basis $g_1, \dots, g_t$ of $\mathcal{A}$ whose first r_i elements form a basis of $\mathcal{A}_i$. The change of basis is an invertible matrix B_i over $\mathbb{F}_q$, so

$$\mathcal{W}_g(X) = \mathcal{W}_f(X) B_i, \qquad \det \mathcal{W}_g(X) = \det(B_i) D(X).$$

Since $\det(B_i)$ is a nonzero constant, this preserves every root and its multiplicity.

For $0 \le u \le s - t$, every entry in the first r_i columns of $\mathcal{W}_g$ is zero at $X = \alpha_i \gamma^u$: these entries evaluate polynomials in $\mathcal{A}_i$ at the block points with exponents $u, \dots, u + t - 1$. We can therefore factor $X - \alpha_i \gamma^u$ from each of these r_i columns. Thus $\det \mathcal{W}_g$, and hence D, has a root of multiplicity at least r_i there:

$$(X - \alpha_i \gamma^u)^{r_i} \mid D(X) \qquad (0 \le u \le s - t).$$

These points are distinct across all blocks. Counting roots with multiplicity therefore gives

$$(s - t + 1) \sum_{i=1}^{n} r_i \le \deg D \le t(k-1).$$

Since $t \le r$ and $R = k/(sn)$,

$$\frac{1}{n} \sum_{i=1}^{n} r_i \le \frac{t(k-1)}{n(s-t+1)} \le \frac{sR}{s-r+1} \dim \mathcal{A}.$$

The trivial bound $r_i \le \dim \mathcal{A}$ gives the truncation at 1 and the case $r > s$. □

For $1 \le r \le s$, FRS therefore avoids every r-LCL profile with $R_*(V_1, \dots, V_n) > sR/(s - r + 1)$. When $r \ll s$, this cutoff is close to R.